

Vinyl Crate

Project Documentation

Fundamentals of Web Technologies

Individual project

Author: **Mehmet Demirci**

Repository: github.com/mdemrc/vinyl-crate

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1. Introduction

Vinyl Crate is a web application for vinyl record collectors. After creating an account, members browse a catalogue of albums, filter it by genre, decade or artist, and open any album to see its full tracklist together with the reviews other members have written.

Every member can rate albums from one to five stars, mark other members' reviews as helpful, and build a personal “crate” — their own record collection — split between titles they already own and a wishlist of titles they still want. Administrators maintain the catalogue and the community through a dedicated panel where they add albums, upload cover art, edit tracklists, manage genres and manage user accounts.

The project follows the architecture of the course laboratory application. It is written in PHP with a MySQL database and uses jQuery for its asynchronous features, so it runs on a standard Apache (XAMPP) server with no build step.

2. Project type and division of work

This is an individual project. All of the work — database design, back end, front end and graphic styling — was carried out by the author, Mehmet Demirci.

3. Source code repository

The complete source code is available on GitHub: <https://github.com/mdemrc/vinyl-crate>

The repository also contains a README with setup instructions and the test accounts: an administrator (admin / admin123) and several regular users (demo / demo123, plus alex, sam and nina with the same name123 pattern).

4. Functionality

Accounts and access

- **User accounts:** registration, login and logout, with passwords stored as hashes (password_hash / password_verify)
- **Roles:** regular members and administrators, with admin-only pages protected by a guard

Catalogue and navigation

- **Browse:** a responsive grid of albums showing cover art, genre, release year and average rating
- **Search:** live phrase search by album title or artist, updating without a page reload
- **Filtering:** by genre (drop-down list), by decade (links) and by artist (links), plus sorting by newest, rating, year or title
- **Quick navigation:** a genre badge or an artist name links straight to the catalogue filtered by that value

Album management - full CRUD (administrators)

- **Create / edit / delete:** administrators create, edit and delete albums
- **Image upload:** cover art is uploaded as a real file, validated by image type and size, and stored on the server
- **Tracklist editor:** a dynamic editor adds or removes track rows, saved as related rows in a separate table

Records and social features

- **Tracklists:** each album shows its ordered tracklist with track lengths and total running time
- **Reviews:** a 1-5 star rating with a comment; one review per user per album, which can be edited
- **Helpful votes:** members mark other people's reviews as helpful, and reviews are ordered by how helpful they are
- **Personal crate:** members mark albums as owned or wishlist and switch between the two instantly

Administration panel

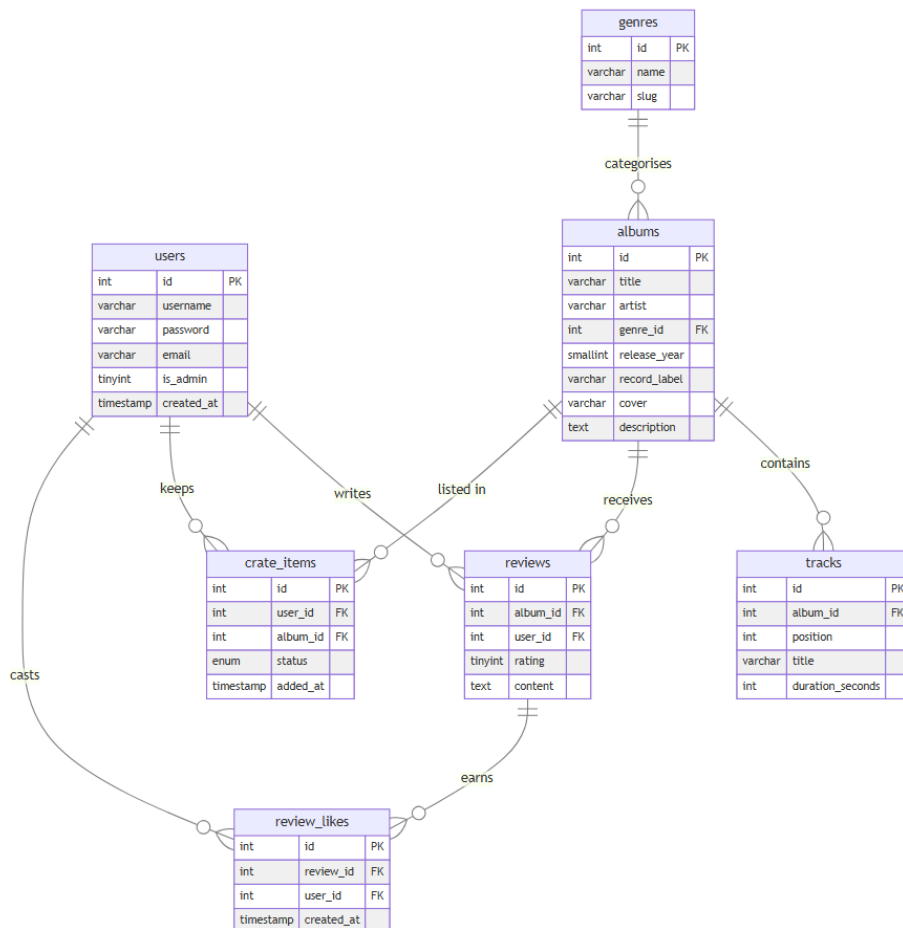
- **Dashboard:** catalogue statistics and a management table of all albums
- **Genres:** add and delete genres
- **Users:** list members with their activity, promote or demote administrators and delete accounts, with protection against an admin locking themselves out
- **Moderation:** delete any member's review directly from the album page

Asynchronous (AJAX) features

- **Without reload:** live search, the owned/wishlist crate toggle, marking a review helpful and review deletion all use asynchronous requests

5. Database schema

The database contains seven related tables. The diagram below shows the tables, their columns and the foreign-key relationships between them.



Entity-relationship diagram of the vinyl_crate database

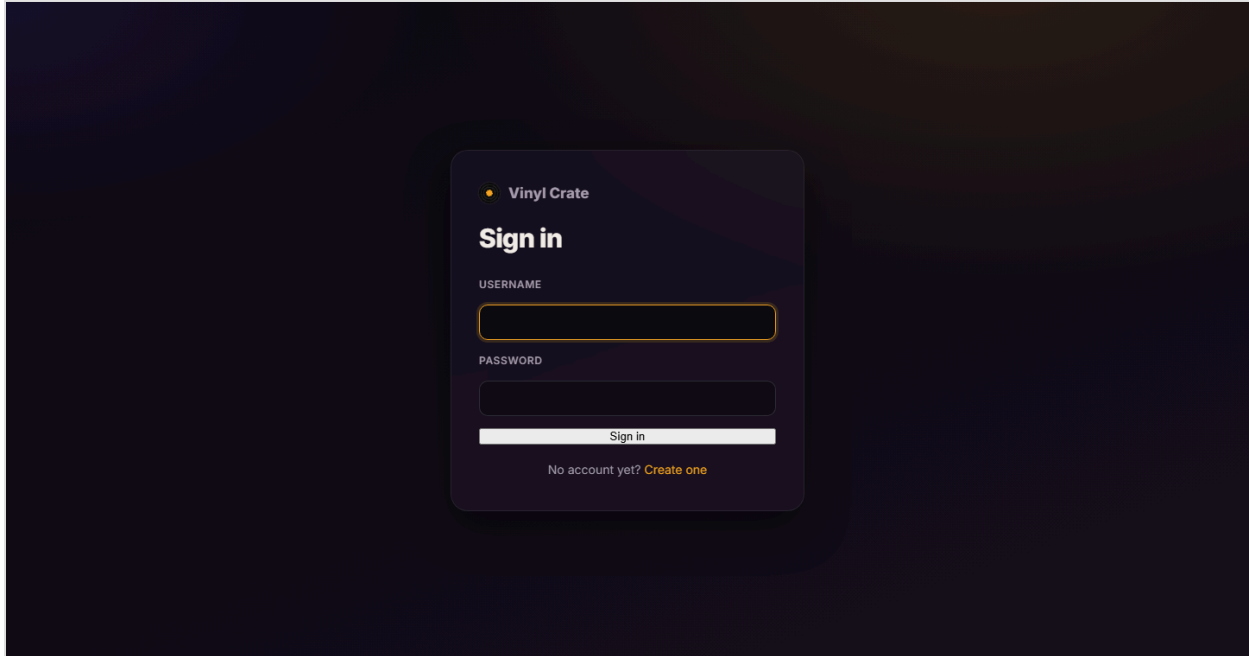
- **users**: user accounts and the is_admin flag
- **genres**: album genres; referenced by albums
- **albums**: the catalogue - each album belongs to one genre
- **tracks**: the tracklist of an album (one album has many tracks)
- **reviews**: one rating and comment per user per album
- **crate_items**: a member's owned and wishlist entries
- **review_likes**: which members marked which reviews as helpful (one vote per user per review)

Deleting an album cascades to its tracks, reviews and crate entries; deleting a review removes its helpful votes; deleting a genre leaves its albums in place (their genre is set to null).

6. Screenshots

Sign in

The entry screen. Visitors sign in or follow the link to create a new account; both forms share the application's dark vinyl theme.



Browse the catalogue

The main catalogue: a grid of album cards with cover art, genre badges, year and average star rating, above the search box, genre/sort drop-downs and decade links.

Vinyl Crate

BrowseMy CrateMy ReviewsLog out (demo)

Browse the crate

All genresNewest first

DECADES: All1950s1960s1970s1980s1990s2010s



Abbey Road
The Beatles

Rock 1969

★★★★★ 5.0 (1)



Innervisions
Stevie Wonder

Soul 1973

No reviews yet



Kind of Blue
Miles Davis

Jazz 1959

★★★★★ 5.0 (1)



OK Computer
Radiohead

Rock 1997

★★★★★ 4.0 (1)



Random Access Memories
Daft Punk

Electronic 2013

★★★★★ 4.0 (1)



Rumours
Fleetwood Mac

Rock 1977

No reviews yet



The Dark Side of the Moon
Pink Floyd

Rock 1973

★★★★★ 5.0 (2)



Thriller
Michael Jackson

Pop 1982

★★★★★ 4.0 (1)



To Pimp a Butterfly
Kendrick Lamar

Hip-Hop 2015

★★★★★ 5.0 (1)



What's Going On
Marvin Gaye

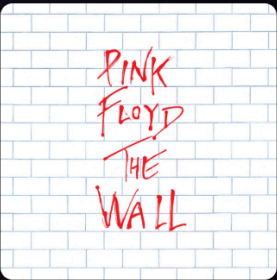
Soul 1971

No reviews yet

Album details

An album page with its cover (a vinyl record peeks out behind it), metadata, the full tracklist with running time, the star-rating review form and the reviews, each of which other members can mark as helpful and an administrator can delete with the corner control.

← Back to the crate



The Dark Side of the Moon

by Pink Floyd

Rock 1970s 1973 Harvest

★★★★★ 4.8/5 from 4 reviews

Pink Floyd's 1973 landmark: one continuous suite on time, money and madness, wrapped in Alan Parsons' pristine engineering.

✓ In collection Add to wishlist Edit album Delete

Tracklist

9 tracks · 39:18

1	Speak to Me	1:30
2	Breathe (In the Air)	2:43
3	On the Run	3:36
4	Time	7:01
5	The Great Gig in the Sky	4:36
6	Money	6:22
7	Us and Them	7:41
8	Brain Damage	3:46
9	Eclipse	2:03

Your review

★★★★★

A timeless masterpiece. The seamless flow from one track to the next still gives me chills on a good turntable.

Update review

Reviews

- admin (you)

2026-06-09 13:54:39

★★★★★

A timeless masterpiece. The seamless flow from one track to the next still gives me chills on a good turntable.

👍 4 found this helpful
- demo

2026-06-09 13:54:39

★★★★★

My desert-island record. The original pressing sounds incredible on vinyl.

👍 Helpful (2)
- alex

2026-06-09 13:54:39

★★★★★

Headphones, lights off, side one through to 'Eclipse'. Nothing else sounds like it.

👍 Helpful (2)
- sam

2026-06-09 13:54:39

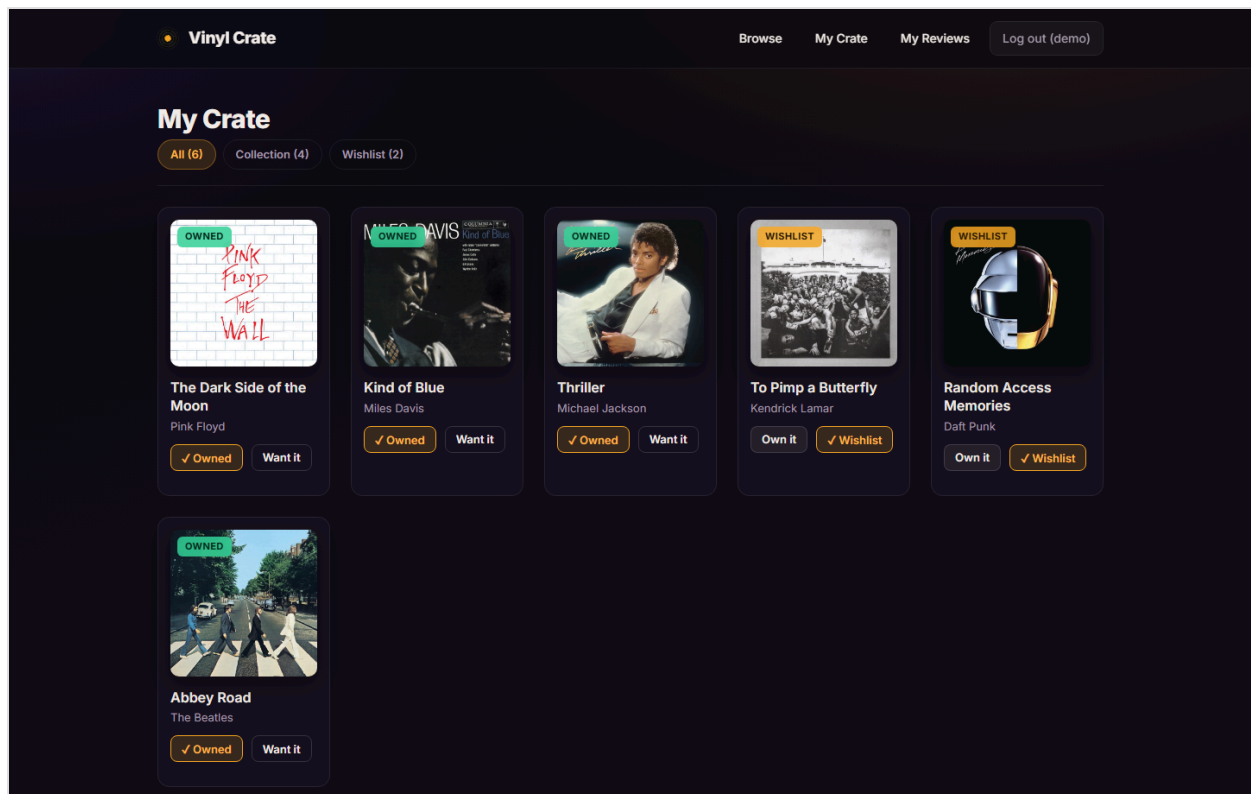
★★★★★

A bit overplayed, but the engineering still holds up better than records made fifty years later.

👍 Helpful (0)

My Crate

The member's personal collection, filtered into the whole crate, owned records and the wishlist, with a live owned/wishlist toggle on each card.



My Reviews

Every review written by the member, with its helpful count, a link back to the album, and inline edit and delete without reloading the page.

Vinyl Crate

[Browse](#)[My Crate](#)[My Reviews](#)[Log out \(demo\)](#)

My Reviews



The Dark Side of the Moon · Pink Floyd
2026-06-09 10:24:44
My desert-island record. The original pressing sounds incredible on vinyl.
👍 2 helpful
[Edit](#) [Delete](#)



Thriller · Michael Jackson
2026-06-09 10:24:44
Pop perfection. Side A is flawless; side B slows down a touch but it is still essential.
👍 1 helpful
[Edit](#) [Delete](#)



Random Access Memories · Daft Punk
2026-06-09 10:24:44
Lush analog production. 'Get Lucky' is the hit, but 'Giorgio' is the heart of the record.
👍 0 helpful
[Edit](#) [Delete](#)



Abbey Road · The Beatles
2026-06-09 10:24:44
The side-two medley is one of the greatest sequences ever committed to wax.
👍 3 helpful
[Edit](#) [Delete](#)



Rumours · Fleetwood Mac
2026-06-09 10:24:44
Heartbreak turned into the catchiest songs of the seventies.
👍 0 helpful
[Edit](#) [Delete](#)

Admin dashboard

The administrator dashboard: catalogue statistics and a management table of every album with edit and delete actions.

Vinyl Crate

[Browse](#)[My Crate](#)[My Reviews](#)[Admin](#)

Log out (admin)

Admin dashboard

Manage users

Manage genres

+ Add album

10

ALBUMS

61

TRACKS

8

GENRES

23

REVIEWS

5

USERS

All albums

	TITLE	ARTIST	GENRE	YEAR	TRACKS	RATING	ACTIONS
	The Dark Side of the Moon	Pink Floyd	Rock	1973	9	4.8/5	Edit Delete
	Kind of Blue	Miles Davis	Jazz	1959	5	4.5/5	Edit Delete
	Thriller	Michael Jackson	Pop	1982	6	4.3/5	Edit Delete
	Rumours	Fleetwood Mac	Rock	1977	6	4.5/5	Edit Delete
	To Pimp a Butterfly	Kendrick Lamar	Hip-Hop	2015	6	4.7/5	Edit Delete
	Random Access Memories	Daft Punk	Electronic	2013	6	4.5/5	Edit Delete
	What's Going On	Marvin Gaye	Soul	1971	5	5.0/5	Edit Delete
	Abbey Road	The Beatles	Rock	1969	6	4.7/5	Edit Delete
	OK Computer	Radiohead	Rock	1997	6	4.5/5	Edit Delete
	Innervisions	Stevie Wonder	Soul	1973	6	4.0/5	Edit Delete

Manage users

User management: every member with their review and collection counts, controls to promote or demote administrators and to delete accounts. The administrator's own row is protected.

Vinyl Crate

[Browse](#)[My Crate](#)[My Reviews](#)[Admin](#)

Log out (admin)

← Back to dashboard

Users

USER	EMAIL	ROLE	JOINED	REVIEWS	RECORDS	ACTIONS
admin (you)	admin@vinylcrate.test	Admin	2026-06-09	5	3	—
demo	demo@vinylcrate.test	User	2026-06-09	5	6	Make admin Delete
alex	alex@vinylcrate.test	User	2026-06-09	5	4	Make admin Delete
sam	sam@vinylcrate.test	User	2026-06-09	4	4	Make admin Delete
nina	nina@vinylcrate.test	User	2026-06-09	4	4	Make admin Delete

Add / edit album

The album form used to create or edit a record, including the cover-image upload field and the dynamic tracklist editor.

The screenshot shows the 'Add a new album' form in the Vinyl Crate application. The form is dark-themed with white text and orange accents. It includes fields for Title, Artist, Genre (a dropdown menu currently showing 'none'), Release Year, Record Label, and a large text area for Description. Below these is a 'COVER IMAGE' section with a 'Choose File' button and 'No file chosen' text. The 'TRACKLIST' section features a '+ Add track' button, a 'Track title' input field, a 'm:ss' duration input field, and a red 'x' button to remove a track. A note below the tracklist says 'Leave a row empty to skip it. Duration format: minutes:seconds (e.g. 4:33)'. At the bottom is a large orange 'Create album' button. The top navigation bar includes 'Browse', 'My Crate', 'My Reviews', 'Admin', and a 'Log out (admin)' button.

Manage genres

Genre management: a form to add a genre and a table listing every genre with its album count and a delete action.

The screenshot shows the 'Manage genres' page in the Vinyl Crate application. It features a 'New genre name' input field and an 'Add genre' button. Below is a table listing existing genres with their slugs, album counts, and delete actions. The top navigation bar is the same as the previous screenshot, with a 'Back to dashboard' link added on the left. The table data is as follows:

NAME	SLUG	ALBUMS	ACTIONS
Classical	classical	0	Delete
Electronic	electronic	1	Delete
Funk	funk	0	Delete
Hip-Hop	hip-hop	1	Delete
Jazz	jazz	1	Delete
Pop	pop	1	Delete
Rock	rock	4	Delete
Soul	soul	2	Delete